# Topic C: common theory, applications and required code

Human Compression project

The common paper *Changes of chamber generators and binary translation groups* supports four complete applications: Janko geometries (C1), halving generalized cubes (C2), Petrials and faithful chamber actions (C3), and regular polyhedra with equal adjacent parameters (C4). The fixed source contracts are L088, N013, L108 and F014. Their forty-row scope table includes the recorded qualifications, retained statements and exclusions; this measurement does not expand that scope.

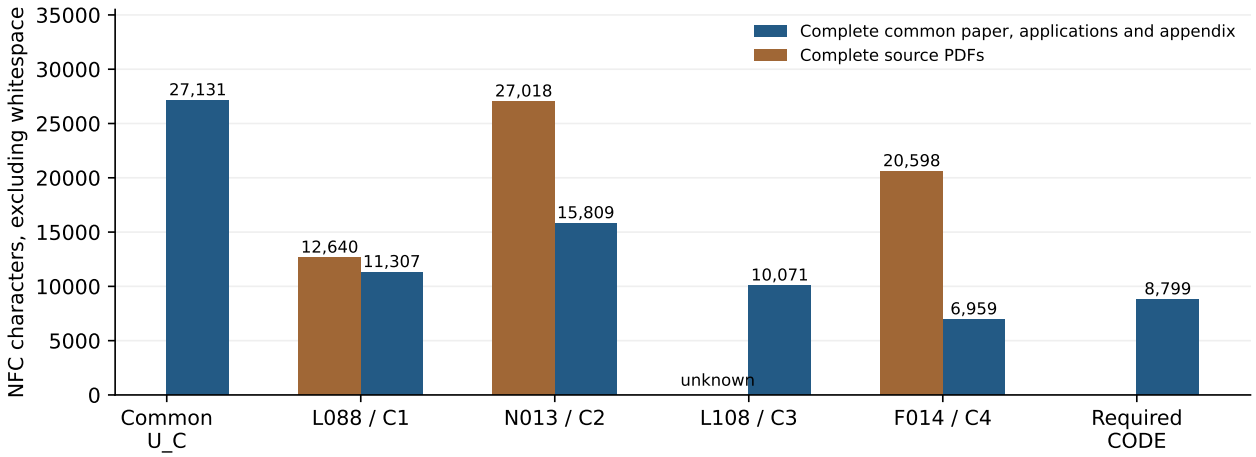
**The whole-topic written comparison.** The whole common theory costs **27,131** characters and the four complete applications cost **44,146**. The written denominator is therefore **71,277**. Three available source PDFs cost 60,256. If $x$ is the unknown original-PDF cost of L108, the written comparison is

$$R_C^{\text{written}} = \frac{x + 60\,256}{71\,277}.$$

The two required programs are also counted in full. Their complete printed appendix costs **8,799**, giving the separately reported comparison

$$R_C^{\text{with programs}} = \frac{x + 60\,256}{80\,076}.$$

Neither comparison has a numerical value until $x$ is known. C3's 10,071 characters remain in both denominators. Its browser text does not replace the missing original PDF.




The unit is NFC characters without whitespace in complete extracted PDF text, after the exact furniture exclusions and visible-glyph corrections. All unique front matter and references remain charged.

**Conditional application comparisons.** After acquiring the *whole common theory and both complete required programs*, the local comparison is the complete source paper divided by its complete application. Both whole-topic denominators pay the common theory once; the program-inclusive one also pays the full appendix once.

| Source / application | Source | Application | Conditional factor |
|---|---|---|---|
| L088 / C1 | 12,640 | 11,307 | 1.118 |
| N013 / C2 | 27,018 | 15,809 | 1.709 |
| L108 / C3 | unknown | 10,071 | unknown |
| F014 / C4 | 20,598 | 6,959 | 2.960 |

**The code is counted once, in a declared representation.** Both GAP files are required mathematical programs and are charged in full. The primary representation is their four-page printed appendix, including its headings and printed glyphs. An alternative uses the complete raw files: 4,458 and 4,783 bytes, with 4,084 and 4,239 NFC nonspace characters. The alternative replaces the appendix; it is never added to it.

| Denominator representation | NFC nonspace cost |
|---|---|
| Common paper and four applications only | 71,277 |
| Written topic plus printed program appendix | 80,076 |
| Written topic plus both raw programs (alternative) | 79,600 |

All three numerators remain $x + 60\,256$. The written comparison is presented first, followed by the required-program-inclusive comparison. The printed appendix is the primary representation of the program cost, not a replacement for the written comparison. The raw-file metrics are summed file by file; no program or diagnostic portion is pruned. There is no cross-topic discount on the common theory.

**What was measured.** The source inputs are three complete physical PDFs, totaling 33 pages. The new side consists of the complete 27-page scientific edition and the four-page required-program appendix. Only exact running furniture and navigation are excluded. Font-aware glyph replacements precede Unicode normalization; no semantic correction, dehyphenation, formula reordering or coalescing of printed delimiter pieces is performed.

The machine-readable record contains all five length units under NFC and NFKC, for raw untrimmed, raw selected and calibrated selected text. Each of these thirty coordinates has the same full denominator boundary and the three declared program representations. Exact conditional fractions appear in the CSV and JSON files. The primary coordinate is the calibrated NFC nonspace count used on page one.

**Scope and retained work.** The Janko application keeps the accepted figure/label qualification. The halving application retains its qualified statement and excludes the specified external-classification comparison and unused semiregular isomorphism. For L108, the unconditional faithful-action assertion is replaced by the exact kernel criterion. F014 uses the fixed first edition and corrected left-action convention; its higher-rank Conjecture 3.7 belongs to the separate question track.

The TomLib subgroup catalogue, group and coset algorithms, Danzer and Pellicer constructions, Coxeter classifications and other stated classical inputs remain attributed and retained. Their expanded proof and catalogue costs are unknown, as are unextracted graphical and formula-layout costs. The extracted-text comparison is not a measurement of total research effort or minimum mathematical complexity.

**Verification and use.** The independently accepted input audit matches 270 PDF metric cells, 54 normalized-text hashes and 20 complete raw-program cells. It checks the embedded fonts and visible glyph conventions. This numerical packet reuses that acceptance and computes only sums, exact fractions and CSV records; it does not repeat the source, GAP or Lean proof work. The 16 scoped Lean declarations and other validation infrastructure remain separately inventoried, without a claim that every application is fully formalized.

Reproduce the arithmetic with `compute.py --out /fresh/directory`; `check.py` verifies the complete boundary, both code representations, all coordinate records and corruption controls. No primary-source text or images are distributed. Independent arithmetic and reader review is separate from this author calculation. No source or question credit is added.